

Introduction to Machine Learning

Machine Learning (ML) is a branch of Artificial Intelligence (AI) that enables computers to learn patterns from data and make predictions or decisions without being explicitly programmed.

Types of Machine Learning

- 1. Supervised Learning:** Learns from labeled data. Examples: House Price Prediction, Spam Detection.
- 2. Unsupervised Learning:** Finds hidden patterns in unlabeled data. Examples: Customer Segmentation, Clustering.
- 3. Reinforcement Learning:** An agent learns by interacting with an environment using rewards and penalties.

Machine Learning Workflow

1. Collect Data
2. Clean and Preprocess Data
3. Feature Engineering
4. Train Model
5. Evaluate Model
6. Deploy Model

Common Algorithms

- Linear Regression
- Logistic Regression
- Decision Trees
- Random Forest
- K-Nearest Neighbors (KNN)
- Naive Bayes
- Support Vector Machine (SVM)
- Neural Networks

Applications

Recommendation Systems, Healthcare, Finance, Fraud Detection, Autonomous Vehicles, Natural Language Processing, Computer Vision, Predictive Maintenance.

Advantages

- Automates decision making
- Improves prediction accuracy
- Learns from data
- Scales to large datasets

Limitations

- Requires quality data
- Can be computationally expensive
- Risk of overfitting
- May be difficult to interpret

Conclusion

Machine Learning is one of the most important technologies in modern software systems. Understanding data, choosing the right algorithm, and evaluating model performance are the foundations of successful ML projects.